

 poolside/**Laguna-S-2.1** 

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  Safetensors
 laguna
 laguna-s-2.1
 vllm
 conversational
 custom\_code
  Eval Results
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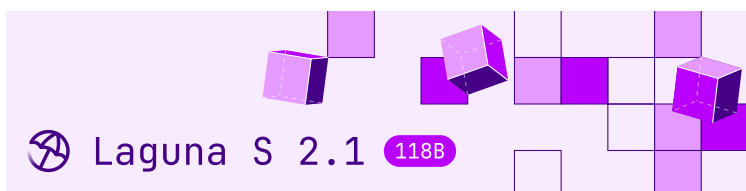
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 **Model card**

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## Laguna S 2.1

Laguna S 2.1 is a 118B total parameter Mixture-of-Experts model with 8B activated parameters per token, designed for agentic coding and long-horizon work. It sits between [Laguna XS 2.1](#) (33B-A3B) and Laguna M.1 (225B-A23B) in the Laguna series and shares the family recipe: a token-choice router with softplus gating over 256 routed experts plus one shared expert, grouped-query attention, and interleaved full/sliding-window attention.

## Highlights

Downloads last month  
**13,285**



 **Safetensors** 

Model size 118B params

Tensor type BF16  Chat template

 Files info

 **Inference Providers** **NEW**

 Text Generation

This model isn't deployed by any Inference Provider.

 14 Ask for provider support

 **Model tree for poolside/Laguna-S...**

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 [poolside/Laguna-S-2.1](#)

 [akhaliq/Laguna-S-2.1](#)

- **Mixed SWA and global attention layout:** 48 layers in a 1:3 global-to-SWA ratio (12 global attention layers, 36 sliding-window layers, window 512), with softplus attention gating and per-layer-type rotary scales
- **1M context:** 1,048,576-token context window
- **Native reasoning support:** interleaved thinking between tool calls, with per-request control via `enable_thinking`
- **Speculative decoding:** a trained [DFlash draft model](#) is available for lower-latency serving
- **Quantized variants:** [FP8](#), [NVFP4](#), [INT4](#) and [GGUF](#)
- **OpenMDW-1.1 license:** Use and modify the model and associated materials freely for commercial and non-commercial purposes ([learn more about OpenMDW](#))

## Model overview

- Number of parameters: 118B total, ~8B activated per token
- Layers: 48 (12 global attention, 36 sliding-window attention)
- Experts: 256 routed (top-10) plus 1 shared expert
- Attention: grouped-query, 8 KV heads, head dim 128; per-head softplus output gating
- Sliding window: 512 tokens
- Context window: 1,048,576 tokens
- Vocabulary: 100,352 tokens (Laguna family tokenizer)

 lyffseba/xai

 bep40/ml-intern

📁 **Collection including** poolside/Lag...

**Laguna S 2.1**  Collection

Our most ca... • 12 items • U... • △ 24

## 📊 Evaluation results ⓘ

datacurve/deep-swe · D...   🏆 40.4

ScaleAI/SWE-bench\_Pro ...   🏆 59.4

SWE-bench/SWE-bench\_Mult...  🏆 78.5

- Modality:text-to-text
- Reasoning:interleaved thinking with preserved thinking

Benchmark results



| Model        | Size     | Terminal-Bench 2.1 | SWE-bench Multilingual | SWE Bench Pro (Public Dataset) |
|--------------|----------|--------------------|------------------------|--------------------------------|
| Laguna S 2.1 | 118B-A8B | 70.2%              | 78.5%                  | 59.4%                          |

| Model               | Size       | Terminal-Bench 2.1 | SWE-bench Multilingual | SWE Bench Pro (Public Dataset) |
|---------------------|------------|--------------------|------------------------|--------------------------------|
| Tencent Hy3         | 295B-A21B  | 71.7%              | 75.8%                  | 57.9%                          |
| Inkling             | 975B-A41B  | 63.8%              | -                      | 54.3%                          |
| Nemotron 3 Ultra    | 550B-A55B  | 56.4%              | 67.7%                  | -                              |
| DeepSeek-V4-Pro Max | 1.6T-A49B  | 64.0%*             | 76.2%                  | 55.4%                          |
| Kimi K3             | 2800B-A50B | 88.3%              | -                      | -                              |
| Qwen 3.7 Max        | -          | 74.5%*             | 78.3%                  | 60.6%                          |
| Muse Spark 1.1      | -          | 80%                | -                      | 61.5%                          |
| Claude Fable 5      | -          | 88%                | -                      | 80.3%                          |

Benchmarks as of 21 July 2026. Laguna S 2.1 in **bold**; a dash (-) marks a benchmark a model was not evaluated on. Scores marked \* are as reported by third parties: Terminal-Bench 2.1 and DeepSWE via Artificial Analysis, SWE Atlas via Scale AI's official leaderboard, and Toolathlon Verified via its official leaderboard. Full evaluation trajectories: [trajectories.poolside.ai](https://trajectories.poolside.ai).

Laguna S 2.1 uses the same laguna architecture as Laguna XS 2.1, so the same engine integrations apply (vLLM, SGLang, Transformers, TRT-LLM, llama.cpp). At 118B parameters the BF16 checkpoint needs multiple GPUs (roughly 236GB of weights); quantized variants reduce this substantially.

## vLLM

```
vllm serve \
  --model poolside/Laguna-S-2.1 \
  --tensor-parallel-size 4 \
  --tool-call-parser poolside_v1 \
  --reasoning-parser poolside_v1 \
  --enable-auto-tool-choice \
  --served-model-name laguna \
  --default-chat-template-kwarg '{"enable.
```

**Optional: speculative decoding with DFlash.** Pair with the [Laguna S2.1 DFlash draft model](#) by adding

```
--speculative-config
'{"model": "poolside/Laguna-S-2.1-
DFlash", "num_speculative_tokens": 7, "method"
: "dflash"}'.
```

## SGLang

```
python -m sglang.launch_server \
  --model-path poolside/Laguna-S-2.1 \
  --tp-size 4 \
  --reasoning-parser poolside_v1 \
  --tool-call-parser poolside_v1 \
  --trust-remote-code
```

## TRT-LLM

```
trtllm-serve poolside/Laguna-S-2.1 --trust-r  
--tool_parser poolside_v1 --reasoning_pa:
```

Note the flag names differ from vLLM's ( --  
tool\_parser, and the reasoning parser is laguna,  
not poolside\_v1).

## llama.cpp

GGUF conversions are available at [poolside/Laguna-S-2.1-GGUF](#). Serve with poolside's llama.cpp fork, branch [laguna](#), which carries full Laguna support including DFlash speculative decoding. (Base Laguna support is also in upstream review: [ggml-org/llama.cpp#25165](#).)

```
git clone --branch laguna https://github.com,  
cd llama.cpp && cmake -B build && cmake --bu  
  
./build/bin/llama-server -m laguna-s-2.1-Q4_I  
  
# with DFlash speculative decoding:  
./build/bin/llama-server -m laguna-s-2.1-Q4_I  
-md laguna-s-2.1-DFlash-BF16.gguf \  
--spec-type draft-dflash --spec-draft-n-ma:
```

## Controlling reasoning

Laguna S 2.1 has native reasoning support and works best with *preserved thinking*: keep reasoning\_content from prior assistant messages in the message history. The model will generally reason before calling tools and between tool calls, and may stop reasoning in follow-up steps if prior thinking blocks are dropped.

Thinking is controlled per request via the chat template:

```
extra_body={"chat_template_kwargs": {"enable
```

or at the server level with `--default-chat-template-kwarg '{"enable_thinking": true}'`. For agentic coding use cases we recommend enabling thinking and preserving reasoning in the message history.

## License

This model is licensed under the [OpenMDW-1.1 License](#).

## Intended and Responsible Use

Laguna S 2.1 is designed for software engineering and agentic coding use cases, and you are responsible for confirming that it is appropriate for your intended application. Laguna S 2.1 is subject to the [OpenMDW-1.1 License](#), and should be used consistently with Poolside's [Acceptable Use Policy](#). We advise against circumventing Laguna S 2.1 safety guardrails without implementing substantially equivalent mitigations appropriate for your use

Please report security vulnerabilities or safety concerns to [security@poolside.ai](mailto:security@poolside.ai).